VEDRICX GLOBAL

Project-Based FEA Industrial Training Program

From Solid Mechanics to Industry-Grade Structural Simulation

Duration: 12 Weeks **Software:** ANSYS Workbench - Mechanical **Fee:** Rs. 6,000 (50% OFF)

Weekly Curriculum

Week 1 — Solid Mechanics Fundamentals
- Stress & strain
- Elastic vs plastic
- Tensile test simulation

Week 2 — Finite Element Representation
- DOF & constraints
- 1D element formulation
- Compression vs bending

Week 3 — 2D/3D Modeling & Meshing
- Plane stress vs strain
- Mesh convergence
- Stress concentration project

Week 4 — Torsional & Shear Behaviour
- Torsional stiffness
- Warping behavior
- Torsion test project

Week 5 — Linear & Nonlinear Analysis
- Material/geometric nonlinearity
- Load stepping
- Plastic tensile analysis

Week 6 — Contact Mechanics
- Bonded/frictional contact
- Penalty methods
- Pin-in-hole assembly

Weeks 7-8 — Dynamics & Harmonic Response
- Modal analysis
- Resonance
- Harmonic response project

Week 9 — Thermal & Thermo-Structural
- Heat conduction
- Thermal-structural coupling

Week 10 — Fatigue & Durability
- S-N curves
- Mean stress correction
- Rotating shaft fatigue

Weeks 11-12 — Capstone Industry Project
- End-to-end FEA
- Mesh convergence & validation
- Complete report & presentation

What You Will Receive

- Complete FEA Technical Report

- ANSYS Project Files
- Engineering Presentation
- Industry-Recognized Certificate

Contact: info@vedricxglobal.com | +91 8439120370